PolyMind Practice Document

Artificial Intelligence

Artificial Intelligence, or AI, is the field of creating computer systems that can perform tasks that normally require human intelligence. Examples include language understanding, image recognition, recommendation systems, and prediction.

Machine Learning

Machine Learning is a subset of AI in which algorithms learn patterns from data instead of being explicitly programmed with every rule. A trained model can use those learned patterns to make predictions on new data.

Neural Networks

Neural networks are computational models made of interconnected layers of artificial neurons. They are widely used in deep learning for tasks such as image classification, speech recognition, and natural language processing.

Natural Language Processing

Natural Language Processing, or NLP, focuses on enabling computers to process and understand human language. Common NLP applications include translation, summarization, sentiment analysis, and question answering.

Retrieval-Augmented Generation

Retrieval-Augmented Generation, commonly called RAG, combines information retrieval with text generation. A RAG system retrieves relevant information from a knowledge source and provides that information to a language model before generating an answer.

Embeddings

An embedding represents information such as text as a numerical vector. Texts with similar meanings tend to have vectors that are close to one another in the embedding space. This makes embeddings useful for semantic search.

FastAPI

FastAPI is a Python framework for building web APIs. It can be used to create backend endpoints through which a frontend application communicates with an AI system.